

NHAI HACKATHON 7.0

Topic :

AuthLens (Secure Offline Facial Verification and Liveness Detection System for Remote Environments)

Submitted by :

Sriza Goel

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Abstract

Identity verification plays a critical role in modern digital governance, transportation infrastructure, public services, and field operations. Most facial verification systems rely heavily on cloud connectivity, making them unsuitable for deployment in remote and low-network regions. In such environments, verification delays can disrupt operations and create security vulnerabilities.

AuthLens is a secure offline facial verification and liveness detection platform designed specifically for remote environments. The system combines facial recognition, challenge-response liveness verification, local audit logging, and lightweight deployment to enable reliable identity verification without requiring continuous internet connectivity.

The solution performs face matching locally, evaluates confidence scores, and triggers intelligent liveness challenges when verification confidence falls into an uncertain range. This approach improves security while maintaining usability in constrained environments.

1. Introduction

Identity verification has become an essential component of modern digital systems. Governments, transportation authorities, border control agencies, and infrastructure operators increasingly depend on biometric authentication to verify individuals quickly and securely.

However, most existing solutions face three major challenges:

Connectivity Dependency

Traditional systems rely on cloud APIs and centralized servers for biometric processing. In remote regions, poor connectivity can make verification impossible.

Security Risks

Static facial verification systems are vulnerable to spoofing attacks using photographs, screenshots, videos, or masks.

High Infrastructure Requirements

Many enterprise-grade verification solutions require expensive hardware and continuous network access.

AuthLens addresses these challenges by providing a lightweight, offline-capable, and secure verification platform.

2. Problem Statement

Remote operations frequently suffer from unstable internet connectivity. Personnel working in highways, toll systems, field offices, border areas, and rural regions often need reliable identity verification systems that can function independently of network infrastructure.

Existing cloud-dependent solutions create several problems:

- Verification delays during network outages.
- Increased operational costs.
- Reduced accessibility in remote regions.
- Greater vulnerability to connectivity failures.
- Limited ability to operate autonomously.

The challenge is to create a highly accurate facial verification and liveness detection system that operates reliably even in zero-network environments.

3. Proposed Solution

AuthLens is designed as a multi-layer identity verification platform that performs facial authentication locally while incorporating challenge-based liveness verification.

The solution follows a three-stage verification pipeline:

Stage 1: Face Capture

The user captures an image through the mobile application.

Stage 2: Facial Verification

The system generates a facial embedding and compares it against locally stored embeddings.

Stage 3: Liveness Validation

When confidence falls within a moderate range, the system requests a dynamic challenge such as:

- Blink
- Turn Head Left
- Turn Head Right

Successful completion confirms the presence of a live individual.

4. Objectives

The primary objectives of AuthLens are:

Security

Prevent unauthorized access through spoofing attacks.

Offline Operation

Enable identity verification without internet connectivity.

Reliability

Provide consistent verification performance across different environments.

Scalability

Support future deployment across large-scale infrastructure networks.

Auditability

Maintain detailed logs of all verification activities.

5. System Architecture

The architecture consists of four major components.

Mobile Application Layer

Responsible for:

- Image capture
- User interaction
- Challenge presentation
- Verification status display

Technology:

- React Native
- Expo

Verification Engine

Responsible for:

- Face embedding generation
- Similarity comparison
- Confidence score calculation

Technology:

- DeepFace
- OpenCV

Liveness Detection Module

Responsible for:

- Blink detection
- Head movement tracking
- Challenge validation

Technology:

- OpenCV
- MediaPipe

Data Storage Layer

Responsible for:

- User records
- Embeddings
- Verification logs

Technology:

- SQLite

6. Workflow

Registration Process

1. User enters identity information.
2. Face image is captured.
3. Facial embedding is generated.
4. Embedding is stored locally.
5. Registration is completed.

Verification Process

1. User captures image.
2. Embedding is generated.
3. Similarity score is calculated.
4. Confidence threshold is evaluated.

Possible outcomes:

Verified

High confidence match.

Challenge Required

Moderate confidence match requiring liveness validation.

Unknown User

Low confidence match.

Challenge Verification

If challenge mode is triggered:

1. Random challenge generated.
2. User performs challenge.
3. Computer vision algorithms validate action.
4. Final verification decision generated.

7. Core Features

Offline Face Recognition

Facial matching occurs locally without cloud dependency.

Benefits:

- Faster processing.
- Reduced latency.
- Increased reliability.
- Better privacy.

Challenge-Based Liveness Detection

Static photographs can deceive simple face recognition systems.

AuthLens introduces active challenges including:

Blink Detection

The system measures eye aspect ratio changes.

Head Movement Detection

The system tracks head orientation and verifies requested movements.

This significantly increases spoof resistance.

Confidence-Based Verification

Instead of binary decisions, AuthLens evaluates similarity confidence.

High Confidence

Immediate approval.

Medium Confidence

Challenge verification required.

Low Confidence

Verification rejected.

This approach improves both security and usability.

Audit Logging

Each verification attempt records:

- User ID
- Confidence score
- Result
- Timestamp

This enables future investigation and compliance auditing.

8. Technology Stack

Frontend

React Native

Backend

Flask

Computer Vision

OpenCV

Used for:

- Image processing.
- Face analysis.
- Challenge verification.

Facial Recognition

DeepFace

Used for:

- Feature extraction.
- Embedding generation.
- Similarity matching.

Database

SQLite

Used for:

- Local storage.
- User management.
- Verification logs.

9. Security Considerations

AuthLens incorporates multiple security mechanisms.

Anti-Spoofing

Challenge-response verification reduces photo attacks.

Local Processing

Sensitive biometric data remains on the device.

Confidence Thresholding

Reduces false acceptance rates.

Audit Trails

All verification events are recorded for accountability.

10. Advantages

Works Without Internet

Designed specifically for remote environments.

Fast Verification

Local processing reduces delays.

Enhanced Security

Multi-stage verification process.

Cost Effective

No cloud infrastructure required.

Privacy Preserving

Data remains within the local environment.

11. Limitations

Current limitations include:

- Performance variations under poor lighting.
- Limited robustness against sophisticated deepfake attacks.
- Dependency on camera quality.
- Initial enrollment quality impacts accuracy.

These limitations can be addressed through future enhancements.

12. Future Enhancements

On-Device AI Inference

Complete removal of backend dependency.

Deepfake Detection

Advanced spoof prevention using texture analysis.

Face Mask Recognition

Support for masked individuals.

Multi-Factor Authentication

Integration with QR codes and digital credentials.

Encryption Layer

Secure storage of biometric embeddings.

Edge AI Optimization

Deployment on low-power hardware devices.

13. Applications

AuthLens can be deployed across various sectors.

Highway and Toll Infrastructure

Personnel verification.

Border Security

Identity validation in remote checkpoints.

Government Field Operations

Offline citizen verification.

Industrial Facilities

Access control in restricted areas.

Disaster Response Teams

Identity verification during emergency operations.

14. Expected Impact

AuthLens demonstrates how modern biometric verification can be adapted for environments with limited connectivity while maintaining security and reliability.

By combining offline facial verification, challenge-based liveness detection, and local audit logging, the platform provides a practical solution for secure identity authentication in remote operational settings.

The system offers a balance between usability, security, and deployability, making it suitable for future expansion into large-scale public infrastructure and government applications.

Conclusion

AuthLens is a secure, lightweight, and offline-capable identity verification platform designed for remote environments. Through facial recognition, challenge-based liveness detection, confidence-driven decision making, and comprehensive audit logging, the system addresses key limitations of conventional cloud-dependent verification systems.

The project demonstrates the feasibility of delivering reliable biometric authentication in low-connectivity scenarios while maintaining security, privacy, and operational efficiency. With future enhancements such as on-device AI inference and advanced anti-spoofing mechanisms, AuthLens has the potential to evolve into a robust next-generation identity verification framework for real-world deployment.